Homework 5-SSIE 553

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Spring 2026

Due March 23, 2026 by 11:59pm through Brightspace

Question 1[10pts each]: Determine the dual of the following Linear Programming Problems:

i

$$\text{Max} \quad 2x_1 + x_2$$

Subject to:

$$-x_1 + x_2 \leq 1 \tag{1}$$

$$x_1 + x_2 \leq 3 \tag{2}$$

$$x_1 - 2x_2 \leq 4 \tag{3}$$

$$x_1, x_2 \geq 0 \tag{4}$$

ii

$$\text{Min} \quad y_1 - y_2$$

Subject to:

$$2y_1 + y_2 \geq 4 \tag{5}$$

$$y_1 + y_2 \geq 1 \tag{6}$$

$$y_1 + 2y_2 \geq 3 \tag{7}$$

$$y_1, y_2 \geq 0 \tag{8}$$

iii

$$\text{Max} \quad 4x_1 - x_2 + 2x_3$$

Subject to:

$$x_1 + x_2 \leq 5 \tag{9}$$

$$2x_1 + x_2 \leq 7 \tag{10}$$

$$2x_2 + x_3 \geq 6 \tag{11}$$

$$x_1 + x_3 = 4 \tag{12}$$

$$x_1 \geq 0 \quad x_2, x_3 \quad urs \tag{13}$$

iv

$$
\begin{aligned}
&\text{Min} \;\; 4y_1 + 2y_2 - y_3 \\
&\text{Subject to:} \\
&y_1 + 2y_2 \le 6 && (14) \\
&y_1 - y_2 + 2y_3 = 8 && (15) \\
&y_1 \ge 0 \;\; y_2, y_3 \;\; urs && (16)
\end{aligned}
$$

Question 2[10pts]: Consider the following problem

$$
\begin{aligned}
&\text{Max} \;\; 2x_1 + 3x_2 + 6x_3 && (17) \\
&\textit{subject to:} && (18) \\
&x_1 + x_3 \le 3 && (19) \\
&x_2 + x_3 \le 5 && (20) \\
&x_1, x_2, x_3 \ge 0 && (21)
\end{aligned}
$$

i Construct the dual problem.

ii Solve the dual problem graphically. Use this solution to identify the shadow prices of the constraints in the primal problem.

Question 3 [15pts]: Consider the following problem

$$
\begin{aligned}
&\text{Max} \;\; x_1 + 2x_2 && (22) \\
&\textit{subject to:} && (23) \\
&-x_1 + x_2 \le -2 && (24) \\
&4x_1 + x_2 \le 4 && (25) \\
&x_1, x_2 \ge 0 && (26)
\end{aligned}
$$

i Demonstrate graphically that this problem has no feasible solutions.

ii Construct the dual problem.

iii Demonstrate graphically that the dual problem is unbounded.

Question 4[20pts]: Consider the following problem

$$
\begin{aligned}
&\text{Max} \;\; 2x_1 + 5x_2 + 4x_3 && (27) \\
&\textit{subject to:} && (28) \\
&x_1 + 2x_2 + 4x_3 \le 10 && (29) \\
&3x_1 + 3x_2 + 2x_3 \le 10 && (30) \\
&x_1, x_2, x_3 \ge 0 && (31)
\end{aligned}
$$

i Construct the dual problem.

ii Use the weak duality theorem to demonstrate that the optimal value of objective function cannot exceed 25.

iii Derive the basic solution and objective function value when x_2 and x_3 are basic variables. Simultaneously derive the corresponding dual solution without explicitly solving the dual. Then draw your conclusions about whether these two basic solutions are optimal for their respective problems using complimentary slackness conditions.

Question 5[15pts]: Company X can manufacture three types of candy bar. Each candy bar consists of sugar and chocolate. Total of 50oz and 100oz of chocolate are available. After defining x_i to be number of Type i candy bars manufactured, Company X should solve the following LP:

$$\text{Max}\, z = 3x_1 + 7x_2 + 5x_3 \;\; \text{(Total Profit in cents)} \tag{32}$$

subject to

$$x_1 + x_2 + x_3 \leq 50 \;\; \text{(Sugar Capacity Constraint)} \tag{33}$$

$$2x_1 + 3x_2 + x_3 \leq 100 \;\; \text{(Chocolate Capacity Constraint)} \tag{34}$$

$$x_i \geq 0 \;\; \forall i = \{1, 2, 3\} \tag{35}$$

After solving the LP, Company X determined the set of optimal basic variables as $\{x_3, x_2\}$ and

$$B^{-1} = \begin{bmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

i Determine the shadow prices for each constraint and give their logical interpretation.

ii How much maximum company X would be willing to pay for 1 unit extra of Sugar.

iii How much maximum company X would be willing to pay for 1 unit extra of Chocolate.